

Quantum Mechanics in Three Dimensions

Quantum Mechanics

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1 Schrödinger Equation in Spherical Coordinates

The time-dependent Schrödinger equation in three dimensions is:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi + V\Psi. \quad (1)$$

The probability density is $|\Psi(\mathbf{r}, t)|^2$ with normalization $\int |\Psi|^2 d^3r = 1$. For time-independent potentials, stationary states are:

$$\Psi_n(\mathbf{r}, t) = \psi_n(\mathbf{r}) e^{-iE_n t/\hbar}, \quad (2)$$

where ψ_n satisfies the time-independent Schrödinger equation:

$$-\frac{\hbar^2}{2m} \nabla^2 \psi_n + V\psi_n = E_n \psi_n. \quad (3)$$

1.1 Separation of Variables

For spherically symmetric potentials $V(r)$, use spherical coordinates (r, θ, ϕ) . The Laplacian is:

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2}{\partial \phi^2}. \quad (4)$$

Assume separable solutions $\psi(r, \theta, \phi) = R(r)Y(\theta, \phi)$. Substituting into Eq. (3) and separating yields:

$$\frac{1}{R} \frac{d}{dr} \left(r^2 \frac{dR}{dr} \right) - \frac{2mr^2}{\hbar^2} [V(r) - E] = l(l+1), \quad (5)$$

$$\frac{1}{Y} \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial Y}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2 Y}{\partial \phi^2} \right] = -l(l+1). \quad (6)$$

The constant $l(l+1)$ is chosen for convenience.

1.2 Angular Equation

Further separation $Y(\theta, \phi) = \Theta(\theta)\Phi(\phi)$ gives:

$$\frac{1}{\Phi} \frac{d^2 \Phi}{d\phi^2} = -m^2, \quad (7)$$

$$\sin \theta \frac{d}{d\theta} \left(\sin \theta \frac{d\Theta}{d\theta} \right) + [l(l+1) \sin^2 \theta - m^2] \Theta = 0. \quad (8)$$

The ϕ equation yields $\Phi(\phi) = e^{im\phi}$. Single-valuedness requires m integer: $m = 0, \pm 1, \pm 2, \dots$. The θ equation is solved by associated Legendre functions:

$$\Theta(\theta) = AP_l^m(\cos \theta). \quad (9)$$

The quantum numbers satisfy: $l = 0, 1, 2, \dots$ and $m = -l, -l+1, \dots, l$.

1.3 Spherical Harmonics

The normalized angular wave functions are spherical harmonics:

$$Y_l^m(\theta, \phi) = \epsilon \sqrt{\frac{2l+1}{4\pi} \frac{(l-|m|)!}{(l+|m|)!}} e^{im\phi} P_l^m(\cos \theta), \quad (10)$$

where $\epsilon = (-1)^m$ for $m \geq 0$ and $\epsilon = 1$ for $m \leq 0$. They satisfy orthonormality:

$$\int_0^{2\pi} \int_0^\pi [Y_l^m(\theta, \phi)]^* Y_{l'}^{m'}(\theta, \phi) \sin \theta d\theta d\phi = \delta_{ll'} \delta_{mm'}. \quad (11)$$

First few spherical harmonics are given in Table 4.2 of the source text.

1.4 Radial Equation

Define $u(r) = rR(r)$. The radial equation becomes:

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[V(r) + \frac{\hbar^2}{2m} \frac{l(l+1)}{r^2} \right] u = Eu. \quad (12)$$

The normalization condition is $\int_0^\infty |u|^2 dr = 1$. The effective potential includes a centrifugal term $\hbar^2 l(l+1)/(2mr^2)$.

1.5 Infinite Spherical Well Example

For $V(r) = 0$ ($r < a$) and $V(r) = \infty$ ($r > a$), the radial equation becomes:

$$\frac{d^2 u}{dr^2} = \left[\frac{l(l+1)}{r^2} - k^2 \right] u, \quad k = \frac{\sqrt{2mE}}{\hbar}. \quad (13)$$

The solutions are spherical Bessel functions:

$$u(r) = A r j_l(kr) + B r n_l(kr), \quad (14)$$

with j_l finite at origin and n_l singular. For normalizability $B = 0$. The boundary condition $j_l(ka) = 0$ gives allowed $k = \beta_{nl}/a$, so:

$$E_{nl} = \frac{\hbar^2 \beta_{nl}^2}{2ma^2}. \quad (15)$$

Each energy level has degeneracy $2l+1$.

2 The Hydrogen Atom

For hydrogen (proton at origin, electron of charge $-e$), the Coulomb potential is:

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}. \quad (16)$$

The radial equation (Eq. 12) becomes:

$$-\frac{\hbar^2}{2m} \frac{d^2 u}{dr^2} + \left[-\frac{e^2}{4\pi\epsilon_0 r} + \frac{\hbar^2}{2m} \frac{l(l+1)}{r^2} \right] u = Eu. \quad (17)$$

2.1 Radial Wave Functions

For bound states ($E < 0$), define:

$$\kappa = \frac{\sqrt{-2mE}}{\hbar}, \quad \rho = \kappa r, \quad \rho_0 = \frac{me^2}{2\pi\epsilon_0\hbar^2\kappa}. \quad (18)$$

Equation (17) becomes:

$$\frac{d^2u}{d\rho^2} = \left[1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2} \right] u. \quad (19)$$

Asymptotic behavior: $u \sim \rho^{l+1}$ as $\rho \rightarrow 0$, $u \sim e^{-\rho}$ as $\rho \rightarrow \infty$. Let:

$$u(\rho) = \rho^{l+1} e^{-\rho} v(\rho). \quad (20)$$

Substituting gives a power series for $v(\rho)$. The recursion relation is:

$$a_{j+1} = \frac{2(j+l+1) - \rho_0}{(j+1)(j+2l+2)} a_j. \quad (21)$$

For normalizability, the series must terminate. This requires:

$$\rho_0 = 2n, \quad n = j_{\max} + l + 1. \quad (22)$$

The principal quantum number n satisfies $n = 1, 2, 3, \dots$ and $l = 0, 1, \dots, n-1$.

2.2 Energy Levels

The allowed energies are given by the Bohr formula:

$$E_n = -\frac{m}{2\hbar^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 \frac{1}{n^2} = \frac{E_1}{n^2}, \quad n = 1, 2, 3, \dots \quad (23)$$

where $E_1 = -13.6$ eV. The degeneracy of level E_n is n^2 .

2.3 Bohr Radius

$$a \equiv \frac{4\pi\epsilon_0\hbar^2}{me^2} = 0.529 \times 10^{-10} \text{ m}. \quad (24)$$

The ground state wave function ($n = 1, l = 0, m = 0$) is:

$$\psi_{100}(r, \theta, \phi) = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}. \quad (25)$$

Higher states are expressed in terms of associated Laguerre polynomials:

$$\psi_{nlm}(r, \theta, \phi) = \sqrt{\left(\frac{2}{na} \right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3}} e^{-r/na} \left(\frac{2r}{na} \right)^l L_{n-l-1}^{2l+1} \left(\frac{2r}{na} \right) Y_l^m(\theta, \phi). \quad (26)$$

2.4 Spectrum of Hydrogen

Transitions between states produce photons of energy:

$$E_\gamma = E_i - E_f = -13.6 \text{ eV} \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right). \quad (27)$$

The Rydberg formula for wavelengths is:

$$\frac{1}{\lambda} = R \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right), \quad R = \frac{m}{4\pi c \hbar^3} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 = 1.097 \times 10^7 \text{ m}^{-1}. \quad (28)$$

Series: Lyman ($n_f = 1$, UV), Balmer ($n_f = 2$, visible), Paschen ($n_f = 3$, IR).

3 Angular Momentum

The orbital angular momentum operator is $\mathbf{L} = \mathbf{r} \times \mathbf{p} = \frac{\hbar}{i}(\mathbf{r} \times \nabla)$.

3.1 Commutation Relations

$$[L_x, L_y] = i\hbar L_z, \quad [L_y, L_z] = i\hbar L_x, \quad [L_z, L_x] = i\hbar L_y, \quad (29)$$

$$[L^2, L_x] = [L^2, L_y] = [L^2, L_z] = 0. \quad (30)$$

Thus L^2 and (say) L_z are compatible observables.

3.2 Eigenvalues

From the commutation relations alone, using ladder operators $L_\pm = L_x \pm iL_y$, we find:

$$L^2 Y_l^m = \hbar^2 l(l+1) Y_l^m, \quad (31)$$

$$L_z Y_l^m = \hbar m Y_l^m, \quad (32)$$

with $l = 0, 1, 2, \dots$ and $m = -l, \dots, l$. The spherical harmonics are the eigenfunctions.

3.3 Raising and Lowering Operators

$$L_\pm Y_l^m = \hbar \sqrt{l(l+1) - m(m \pm 1)} Y_l^{m \pm 1}. \quad (33)$$

4 Spin

Elementary particles carry intrinsic angular momentum called spin, with commutation relations identical to orbital angular momentum:

$$[S_x, S_y] = i\hbar S_z, \quad [S_y, S_z] = i\hbar S_x, \quad [S_z, S_x] = i\hbar S_y. \quad (34)$$

Eigenvalues: $S^2 |sm\rangle = \hbar^2 s(s+1) |sm\rangle$, $S_z |sm\rangle = \hbar m |sm\rangle$, with $s = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ and $m = -s, \dots, s$. For a given particle species, s is fixed.

4.1 Spin 1/2

For $s = 1/2$, the basis states are spin up (χ_+) and spin down (χ_-). The Pauli spin matrices $\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ satisfy $S = (\hbar/2)\boldsymbol{\sigma}$:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (35)$$

The general spinor is $\chi = \begin{pmatrix} a \\ b \end{pmatrix}$ with $|a|^2 + |b|^2 = 1$.

4.2 Electron in a Magnetic Field

For a uniform field $\mathbf{B} = B_0 \hat{k}$, the Hamiltonian is $H = -\gamma B_0 S_z$. The Larmor precession frequency is $\omega = \gamma B_0$, and $\langle \mathbf{S} \rangle$ precesses about the field.

4.3 Addition of Angular Momenta

For two spins s_1 and s_2 , the total spin quantum number s takes values:

$$s = s_1 + s_2, s_1 + s_2 - 1, \dots, |s_1 - s_2|. \quad (36)$$

The combined states are given by Clebsch-Gordan coefficients:

$$|sm\rangle = \sum_{m_1+m_2=m} C_{m_1 m_2 m}^{s_1 s_2 s} |s_1 m_1\rangle |s_2 m_2\rangle. \quad (37)$$

For two spin-1/2 particles:

$$|1\ 1\rangle = \uparrow\uparrow, \quad (38)$$

$$|1\ 0\rangle = \frac{1}{\sqrt{2}}(\uparrow\downarrow + \downarrow\uparrow), \quad (39)$$

$$|1\ -1\rangle = \downarrow\downarrow, \quad (40)$$

$$|0\ 0\rangle = \frac{1}{\sqrt{2}}(\uparrow\downarrow - \downarrow\uparrow). \quad (41)$$

These are the triplet ($s = 1$) and singlet ($s = 0$) states.

5 Key Results Summary

$$\text{Hydrogen energies: } E_n = -\frac{13.6\text{ eV}}{n^2}. \quad (42)$$

$$\text{Bohr radius: } a = \frac{4\pi\epsilon_0\hbar^2}{me^2} = 0.529 \times 10^{-10}\text{ m}. \quad (43)$$

$$\text{Angular momentum: } L^2 = \hbar^2 l(l+1), \quad L_z = \hbar m. \quad (44)$$

$$\text{Spin 1/2: } S = \frac{\hbar}{2}\boldsymbol{\sigma}, \quad \sigma_i \sigma_j = \delta_{ij} + i \sum_k \epsilon_{ijk} \sigma_k. \quad (45)$$

$$\text{Addition of spins: } s = |s_1 - s_2|, \dots, s_1 + s_2. \quad (46)$$